

Target validation paths — chondrosarcoma-mets-acetabular-fx-id h-unknown-y34r

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

Two orders decide most of what this case can become, and both ride the archival diagnostic block: IDH1/IDH2 sequencing, and a comprehensive tumour DNA plus RNA panel read on macrodissected dedifferentiated tissue. A confirmed IDH1 or IDH2 mutation opens an off-label conversation about ivosidenib or olutasidenib rather than a trial seat, because the active chondrosarcoma IDH studies are closed to this histology (the phase 3 ivosidenib trial NCT06127407 enrolls conventional histology only); a wild-type result closes that axis cleanly and leaves the rest of the workup standing. The same panel's TMB, MSI and CDK-pathway readouts decide whether tumour-agnostic pembrolizumab, the abemaciclib study NCT04040205 and TAPUR (NCT02693535) matching are reachable at all. If the panel comes back quiet, which chondrosarcoma genomes often do beyond IDH, the molecular half of this report is exhausted, and the conversation about standard systemic care from there belongs to the treating team, not Libby.

IDH1/IDH2

Order sequencing, not R132H immunohistochemistry. The R132H antibody was raised for glioma, and roughly 40 percent of IDH1-mutant cartilage tumours carry R132C instead ([PMID 21598255](#)), so a stain would read negative in the commonest mutant scenario here. The pre-test probability is high: 71 percent of dedifferentiated chondrosarcomas carried an IDH1/IDH2 mutation in a paired genomic series, against 36 percent of conventional tumours ([PMID 36926116](#)). A targeted hotspot assay answers in 3-10 days; on the comprehensive panel the same answer takes 2-3 weeks. Two sampling rules matter as much as the assay. Macrodissect the dedifferentiated component, and require the report to state which lesion was sampled, because mutant IDH1 is recurrently lost during metastatic evolution ([PMID 41299743](#)) and a call on the acetabular primary may not describe the lung nodules. Plasma ctDNA (two Streck tubes, 7-10 days) is the fallback when the block is exhausted, and the shedding data are favourable in exactly this population: digital PCR found circulating tumour DNA in all six patients with grade 3 or dedifferentiated disease ([PMID 28834325](#)). Whatever comes back, temper the expectations that travel with the word "mutation": the widely quoted ivosidenib response rate belongs to conventional-histology patients, and olutasidenib in dedifferentiated disease read out at a median PFS of 1.5 months.

Dedifferentiated histology

The histology on file is user-stated, and it is the fact that makes cytotoxic chemotherapy worth giving, so hardening it is essential rather than optional. Expert bone-sarcoma pathology review should report the formal WHO diagnosis and the percentage of the specimen occupied by the dedifferentiated, non-cartilaginous component; it runs 5-10 business days on the existing slides and block, no new biopsy. Ancillary subtyping earns its slot because the differential changes the chemotherapy backbone: HEY1-NCOA2 fusion testing on an RNA panel separates mesenchymal chondrosarcoma, and morphology plus the IDH genotype separates chondroblastic osteosarcoma, which would move treatment toward a MAP-type regimen. Two baseline studies stand between her and an anthracycline: a transthoracic echocardiogram with LVEF plus a 12-lead ECG with QTc (1-3 days), and an organ-function panel with creatinine clearance and urinalysis before any ifosfamide or cisplatin.

Molecular landscape

Everything molecular in this case runs off one block, and dedifferentiated chondrosarcoma is biphasic by definition, so the pathology block review with macrodissection instructions and a written tissue-allocation plan comes before anything is cut. A core that sampled only the low-grade cartilage returns a falsely reassuring result

on every assay downstream of it. The comprehensive DNA plus RNA panel (IDH1/IDH2, TP53, CDKN2A/B, TERT promoter, NTRK1-3, RET, BRAF, TMB, MSI, fusions; 2-3 weeks) is the single order that answers the IDH question and every tumour-agnostic one at once. Around it sit the fast reads: four-antibody mismatch-repair IHC (2-5 days off four unstained slides) as the orthogonal check on the panel's MSI call, which would put tumour-agnostic pembrolizumab or dostarlimab in reach if protein loss appears, though MSI-high showed up in only 0.7 percent of sarcomas in the largest sequenced cohort ([PMID 42531431](#)); PD-L1 IHC with the clone and scoring system recorded explicitly, read on the dedifferentiated component, since positivity in this disease is confined to that subtype (9 of 22 dedifferentiated against 0 of 119 conventional, [PMID 27312065](#)); and CD3/CD8 density with tumour HLA class I expression as context for any checkpoint or cell-therapy discussion. A germline predisposition panel is a reflex, ordered only if the tumour panel returns a BRCA1, BRCA2, PALB2 or TP53 variant at an allele fraction suggesting germline origin.

Fitness, fracture and measurable disease

The recorded ECOG of 2 is an intake assumption made from the fracture, not a score, and it decides more than any single molecular result: chemotherapy intensity, and eligibility for every protocol that caps performance status, including the ECOG 0-1 bars on the IACS-6274 study NCT05039801 and the abemaciclib study NCT04040205. Scoring it in clinic is free and same-day; record weight-bearing status, ambulatory aids and analgesic requirement alongside it. Orthopaedic oncology assessment of the periacetabular fracture, with thin-slice bone-algorithm pelvic CT and MRI of the hemipelvis, settles whether fixation precedes or accompanies systemic therapy and starts the surgical-washout clock that trial protocols read. A baseline contrast CT of chest, abdomen and pelvis with thin lung reconstructions establishes RECIST 1.1 measurability, which protocol entry generally requires, and answers whether the bilateral nodule distribution truly forecloses any local pulmonary approach.

HLA class I and the cancer-testis antigens

One blood draw that spends no tumour tissue: allele-level (4-digit) HLA class I typing (1-2 weeks) decides whether any HLA-restricted TCR-T or ImmTAC programme, afamitresgene autoleucel, uzatresgene autoleucel, IMA203, brentafusp or letetresgene autoleucel, can be pursued at all. The antigen stains behind it are reflexes, not parallel orders: MAGE-A4, NY-ESO-1 and PRAME expression are worth measuring only after a documented HLA-A*02:01, because an expression result cannot be read for eligibility without the restricting allele. Hold expectations low even then; in the largest cancer-testis antigen survey of bone and soft tissue tumours, 1 of 13 chondrosarcomas stained PRAME-positive ([PMID 40152485](#)).

Biomarkers measured, but not to decision resolution

This case has no under-resolved results, because it has no results: nothing molecular has been measured at all, and the survey behind this report handed four items forward on exactly that basis rather than because an existing number fell short. HLA class I genotype is simply absent, and it is the one that stacks: MAGE-A4, NY-ESO-1 and PRAME are recorded as not assessable rather than negative, since two gaps sit on top of each other for each of them, an unknown restricting allele and an expression level nobody has stained for. Each of the four has a matching workup row above, and the ordering dependency is the whole story: type first, stain second, and read any expression result on the dedifferentiated component.

Where to order these assays

Assay	Provider	Decision gated	Contact
IDH1 codon 132 and IDH2 codons 140/172 mutation testing by DNA sequencing (comprehensive NGS panel or targeted hotspot assay), not R132H immunohistochemistry	Foundation Medicine (preferred) (FoundationOne CDx)	Ivosidenib or olutasidenib and entry to IDH1-directed trials, each of which requires a documented mutation by sequencing.	test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639
IDH1 codon 132 and IDH2 codons 140/172 mutation testing by DNA sequencing (comprehensive NGS panel or targeted hotspot assay), not R132H immunohistochemistry	Caris Life Sciences (MI Cancer Seek / MI Profile)	Ivosidenib or olutasidenib and entry to IDH1-directed trials, each of which requires a documented mutation by sequencing.	test info · 3600 W Royal Ln, Irving, TX 75063 · 888-979-8669
IDH1 codon 132 and IDH2 codons 140/172 mutation testing by DNA sequencing (comprehensive NGS panel or targeted hotspot assay), not R132H immunohistochemistry	Tempus AI (Tempus xT)	Ivosidenib or olutasidenib and entry to IDH1-directed trials, each of which requires a documented mutation by sequencing.	test info · 600 West Chicago Avenue, Chicago, IL 60654
IDH1 codon 132 and IDH2 codons 140/172 mutation testing by DNA sequencing (comprehensive NGS panel or targeted hotspot assay), not R132H immunohistochemistry	NeoGenomics Laboratories (NeoTYPE Sarcoma Profile)	Ivosidenib or olutasidenib and entry to IDH1-directed trials, each of which requires a documented mutation by sequencing.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 239-768-0600 option 3
IDH1 codon 132 and IDH2 codons 140/172 mutation testing by DNA sequencing (comprehensive NGS panel or targeted hotspot assay), not R132H immunohistochemistry	Mayo Clinic Laboratories (IDH1/IDH2 targeted mutation analysis)	Ivosidenib or olutasidenib and entry to IDH1-directed trials, each of which requires a documented mutation by sequencing.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
Expert bone-sarcoma pathology review of the diagnostic material, reporting the WHO diagnosis and the percentage of the specimen occupied by the dedifferentiated (non-cartilaginous) component	Mayo Clinic Laboratories (preferred) (Consultative bone and soft tissue pathology)	Whether anthracycline-based chemotherapy is the correct first-line choice, and which sarcoma protocols the diagnosis fits.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
Expert bone-sarcoma pathology review of the diagnostic material, reporting the WHO diagnosis and the percentage of the specimen occupied by the dedifferentiated (non-cartilaginous) component	Memorial Sloan Kettering Cancer Center, Department of Pathology and Laboratory Medicine (Pathology consultation service)	Whether anthracycline-based chemotherapy is the correct first-line choice, and which sarcoma protocols the diagnosis fits.	test info · 1275 York Avenue, New York, NY 10065
Expert bone-sarcoma pathology review of the diagnostic material, reporting the WHO diagnosis and the percentage of the specimen occupied by the dedifferentiated (non-cartilaginous) component	MD Anderson Cancer Center, Division of Pathology and Laboratory Medicine (Pathology second-opinion consultation)	Whether anthracycline-based chemotherapy is the correct first-line choice, and which sarcoma protocols the diagnosis fits.	test info · 1515 Holcombe Boulevard, Houston, TX 77030

Expert bone-sarcoma pathology review of the diagnostic material, reporting the WHO diagnosis and the percentage of the specimen occupied by the dedifferentiated (non-cartilaginous) component	Johns Hopkins Department of Pathology (Surgical pathology consultation)	Whether anthracycline-based chemotherapy is the correct first-line choice, and which sarcoma protocols the diagnosis fits.	test info · 600 North Wolfe Street, Baltimore, MD 21287
Comprehensive tumour DNA plus RNA next-generation sequencing panel covering IDH1/IDH2, TP53, CDKN2A/B, TERT promoter, NTRK1-3, RET and BRAF, reporting TMB in mutations per megabase and an MSI call	Foundation Medicine (preferred) (FoundationOne CDx)	Tumour-agnostic options (TRK and RET inhibitors, dabrafenib plus trametinib, pembrolizumab for TMB-H or MSI-H) and genotype-matched trial entry.	test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639
Comprehensive tumour DNA plus RNA next-generation sequencing panel covering IDH1/IDH2, TP53, CDKN2A/B, TERT promoter, NTRK1-3, RET and BRAF, reporting TMB in mutations per megabase and an MSI call	Caris Life Sciences (MI Cancer Seek (whole exome plus whole transcriptome))	Tumour-agnostic options (TRK and RET inhibitors, dabrafenib plus trametinib, pembrolizumab for TMB-H or MSI-H) and genotype-matched trial entry.	test info · 3600 W Royal Ln, Irving, TX 75063 · 888-979-8669
Comprehensive tumour DNA plus RNA next-generation sequencing panel covering IDH1/IDH2, TP53, CDKN2A/B, TERT promoter, NTRK1-3, RET and BRAF, reporting TMB in mutations per megabase and an MSI call	Tempus AI (Tempus xT plus xR)	Tumour-agnostic options (TRK and RET inhibitors, dabrafenib plus trametinib, pembrolizumab for TMB-H or MSI-H) and genotype-matched trial entry.	test info · 600 West Chicago Avenue, Chicago, IL 60654
Comprehensive tumour DNA plus RNA next-generation sequencing panel covering IDH1/IDH2, TP53, CDKN2A/B, TERT promoter, NTRK1-3, RET and BRAF, reporting TMB in mutations per megabase and an MSI call	NeoGenomics Laboratories (NeoTYPE Sarcoma Profile)	Tumour-agnostic options (TRK and RET inhibitors, dabrafenib plus trametinib, pembrolizumab for TMB-H or MSI-H) and genotype-matched trial entry.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 239-768-0600 option 3
Comprehensive tumour DNA plus RNA next-generation sequencing panel covering IDH1/IDH2, TP53, CDKN2A/B, TERT promoter, NTRK1-3, RET and BRAF, reporting TMB in mutations per megabase and an MSI call	Mayo Clinic Laboratories (OncoHeme / solid tumour comprehensive panel)	Tumour-agnostic options (TRK and RET inhibitors, dabrafenib plus trametinib, pembrolizumab for TMB-H or MSI-H) and genotype-matched trial entry.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
HLA class I genotyping at 4-digit (allele-level) resolution by sequence-based typing	Mayo Clinic Laboratories (preferred) (HLA class I high-resolution typing)	Whether HLA-restricted TCR-T and ImmTAC programmes (afamitresgene autoleucel, uzatresgene autoleucel, IMA203, brenetafusp) can be pursued at all.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710

HLA class I genotyping at 4-digit (allele-level) resolution by sequence-based typing	HistoGenetics (NGS-based HLA typing)	Whether HLA-restricted TCR-T and ImmTAC programmes (afamitresgene autoleucel, uzatresgene autoleucel, IMA203, brenetafusp) can be pursued at all.	test info · Ossining, NY
HLA class I genotyping at 4-digit (allele-level) resolution by sequence-based typing	Versiti Diagnostic Laboratories (HLA typing)	Whether HLA-restricted TCR-T and ImmTAC programmes (afamitresgene autoleucel, uzatresgene autoleucel, IMA203, brenetafusp) can be pursued at all.	test info · Milwaukee, WI
HLA class I genotyping at 4-digit (allele-level) resolution by sequence-based typing	Labcorp Oncology (HLA typing)	Whether HLA-restricted TCR-T and ImmTAC programmes (afamitresgene autoleucel, uzatresgene autoleucel, IMA203, brenetafusp) can be pursued at all.	test info · 358 South Main Street, Burlington, NC 27215
HLA class I genotyping at 4-digit (allele-level) resolution by sequence-based typing	Quest Diagnostics (HLA typing)	Whether HLA-restricted TCR-T and ImmTAC programmes (afamitresgene autoleucel, uzatresgene autoleucel, IMA203, brenetafusp) can be pursued at all.	test info · 500 Plaza Drive, Secaucus, NJ 07094
Four-antibody mismatch-repair IHC (MLH1, PMS2, MSH2, MSH6), with MSI by PCR or NGS if any stain is equivocal	NeoGenomics Laboratories (preferred) (MMR IHC panel)	Tumour-agnostic pembrolizumab or dostarlimab in a later line if mismatch-repair protein loss is found.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 239-768-0600 option 3
Four-antibody mismatch-repair IHC (MLH1, PMS2, MSH2, MSH6), with MSI by PCR or NGS if any stain is equivocal	Labcorp Oncology (MMR IHC panel)	Tumour-agnostic pembrolizumab or dostarlimab in a later line if mismatch-repair protein loss is found.	test info · 358 South Main Street, Burlington, NC 27215
Four-antibody mismatch-repair IHC (MLH1, PMS2, MSH2, MSH6), with MSI by PCR or NGS if any stain is equivocal	Quest Diagnostics (MMR IHC panel)	Tumour-agnostic pembrolizumab or dostarlimab in a later line if mismatch-repair protein loss is found.	test info · 500 Plaza Drive, Secaucus, NJ 07094
Four-antibody mismatch-repair IHC (MLH1, PMS2, MSH2, MSH6), with MSI by PCR or NGS if any stain is equivocal	Mayo Clinic Laboratories (Mismatch repair protein IHC)	Tumour-agnostic pembrolizumab or dostarlimab in a later line if mismatch-repair protein loss is found.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
Four-antibody mismatch-repair IHC (MLH1, PMS2, MSH2, MSH6), with MSI by PCR or NGS if any stain is equivocal	ARUP Laboratories (MMR IHC panel)	Tumour-agnostic pembrolizumab or dostarlimab in a later line if mismatch-repair protein loss is found.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 800-522-2787

Ancillary subtyping to separate dedifferentiated chondrosarcoma from chondroblastic osteosarcoma, mesenchymal chondrosarcoma and grade 3 conventional chondrosarcoma: morphology review plus RNA fusion testing for HEY1-NCOA2 and the IDH1/IDH2 genotype	Caris Life Sciences (preferred) (MI Cancer Seek whole transcriptome)	Choice of chemotherapy backbone; a chondroblastic osteosarcoma call would move treatment toward a MAP-type regimen.	test info · 3600 W Royal Ln, Irving, TX 75063 · 888-979-8669
Ancillary subtyping to separate dedifferentiated chondrosarcoma from chondroblastic osteosarcoma, mesenchymal chondrosarcoma and grade 3 conventional chondrosarcoma: morphology review plus RNA fusion testing for HEY1-NCOA2 and the IDH1/IDH2 genotype	Tempus AI (Tempus xR)	Choice of chemotherapy backbone; a chondroblastic osteosarcoma call would move treatment toward a MAP-type regimen.	test info · 600 West Chicago Avenue, Chicago, IL 60654
Ancillary subtyping to separate dedifferentiated chondrosarcoma from chondroblastic osteosarcoma, mesenchymal chondrosarcoma and grade 3 conventional chondrosarcoma: morphology review plus RNA fusion testing for HEY1-NCOA2 and the IDH1/IDH2 genotype	NeoGenomics Laboratories (NeoTYPE Sarcoma Profile with fusion panel)	Choice of chemotherapy backbone; a chondroblastic osteosarcoma call would move treatment toward a MAP-type regimen.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 239-768-0600 option 3
Ancillary subtyping to separate dedifferentiated chondrosarcoma from chondroblastic osteosarcoma, mesenchymal chondrosarcoma and grade 3 conventional chondrosarcoma: morphology review plus RNA fusion testing for HEY1-NCOA2 and the IDH1/IDH2 genotype	Foundation Medicine (FoundationOne CDx)	Choice of chemotherapy backbone; a chondroblastic osteosarcoma call would move treatment toward a MAP-type regimen.	test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639
Ancillary subtyping to separate dedifferentiated chondrosarcoma from chondroblastic osteosarcoma, mesenchymal chondrosarcoma and grade 3 conventional chondrosarcoma: morphology review plus RNA fusion testing for HEY1-NCOA2 and the IDH1/IDH2 genotype	Mayo Clinic Laboratories (Sarcoma fusion gene panel, RNA-based)	Choice of chemotherapy backbone; a chondroblastic osteosarcoma call would move treatment toward a MAP-type regimen.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710

IDH1/IDH2 sequencing performed on macrodissected dedifferentiated tumour, with the sampled lesion documented on the report; consider a metastatic-site sample if an IDH-directed therapy is being pursued	Foundation Medicine (preferred) (FoundationOne CDx)	How far an IDH1-mutant result from the primary can be trusted when treating the metastatic disease.	test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639
IDH1/IDH2 sequencing performed on macrodissected dedifferentiated tumour, with the sampled lesion documented on the report; consider a metastatic-site sample if an IDH-directed therapy is being pursued	Caris Life Sciences (MI Cancer Seek)	How far an IDH1-mutant result from the primary can be trusted when treating the metastatic disease.	test info · 3600 W Royal Ln, Irving, TX 75063 · 888-979-8669
IDH1/IDH2 sequencing performed on macrodissected dedifferentiated tumour, with the sampled lesion documented on the report; consider a metastatic-site sample if an IDH-directed therapy is being pursued	Mayo Clinic Laboratories (Solid tumour NGS with macrodissection)	How far an IDH1-mutant result from the primary can be trusted when treating the metastatic disease.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
PD-L1 IHC with the clone and scoring system recorded explicitly (22C3 pharmDx with CPS, or SP263 per the protocol in play), read on a section containing the dedifferentiated component	NeoGenomics Laboratories (preferred) (PD-L1 IHC (22C3, SP263, 28-8))	Enrichment for checkpoint-inhibitor trials; it does not gate an approved drug in sarcoma.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 239-768-0600 option 3
PD-L1 IHC with the clone and scoring system recorded explicitly (22C3 pharmDx with CPS, or SP263 per the protocol in play), read on a section containing the dedifferentiated component	Labcorp Oncology (PD-L1 IHC)	Enrichment for checkpoint-inhibitor trials; it does not gate an approved drug in sarcoma.	test info · 358 South Main Street, Burlington, NC 27215
PD-L1 IHC with the clone and scoring system recorded explicitly (22C3 pharmDx with CPS, or SP263 per the protocol in play), read on a section containing the dedifferentiated component	Quest Diagnostics (PD-L1 IHC)	Enrichment for checkpoint-inhibitor trials; it does not gate an approved drug in sarcoma.	test info · 500 Plaza Drive, Secaucus, NJ 07094
PD-L1 IHC with the clone and scoring system recorded explicitly (22C3 pharmDx with CPS, or SP263 per the protocol in play), read on a section containing the dedifferentiated component	Mayo Clinic Laboratories (PD-L1 IHC)	Enrichment for checkpoint-inhibitor trials; it does not gate an approved drug in sarcoma.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710

PD-L1 IHC with the clone and scoring system recorded explicitly (22C3 pharmDx with CPS, or SP263 per the protocol in play), read on a section containing the dedifferentiated component	ARUP Laboratories (PD-L1 IHC)	Enrichment for checkpoint-inhibitor trials; it does not gate an approved drug in sarcoma.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 800-522-2787
MAGE-A4 expression by IHC or RT-PCR on the dedifferentiated component, ordered only after HLA-A*02:01 is documented; RNA expression from a whole-transcriptome panel is an acceptable first pass	Caris Life Sciences (preferred) (MI Cancer Seek whole transcriptome (MAGEA4 expression))	MAGE-A4-directed TCR-T referral; afamitresgene autoleucel is synovial-sarcoma labelled, uzatresgene autoleucel is trial-only.	test info · 3600 W Royal Ln, Irving, TX 75063 · 888-979-8669
MAGE-A4 expression by IHC or RT-PCR on the dedifferentiated component, ordered only after HLA-A*02:01 is documented; RNA expression from a whole-transcriptome panel is an acceptable first pass	Tempus AI (Tempus xR RNA expression)	MAGE-A4-directed TCR-T referral; afamitresgene autoleucel is synovial-sarcoma labelled, uzatresgene autoleucel is trial-only.	test info · 600 West Chicago Avenue, Chicago, IL 60654
MAGE-A4 expression by IHC or RT-PCR on the dedifferentiated component, ordered only after HLA-A*02:01 is documented; RNA expression from a whole-transcriptome panel is an acceptable first pass	NeoGenomics Laboratories (Custom immunohistochemistry)	MAGE-A4-directed TCR-T referral; afamitresgene autoleucel is synovial-sarcoma labelled, uzatresgene autoleucel is trial-only.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 239-768-0600 option 3
Plasma ctDNA NGS panel covering IDH1 R132 and IDH2 R140/R172 hotspots	Guardant Health (preferred) (Guardant360 CDx)	Fallback route to the IDH genotype when tissue is inadequate or exhausted.	test info · 3100 Hanover Street, Palo Alto, CA 94304
Plasma ctDNA NGS panel covering IDH1 R132 and IDH2 R140/R172 hotspots	Foundation Medicine (FoundationOne Liquid CDx)	Fallback route to the IDH genotype when tissue is inadequate or exhausted.	test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639
Plasma ctDNA NGS panel covering IDH1 R132 and IDH2 R140/R172 hotspots	Tempus AI (Tempus xF)	Fallback route to the IDH genotype when tissue is inadequate or exhausted.	test info · 600 West Chicago Avenue, Chicago, IL 60654
Plasma ctDNA NGS panel covering IDH1 R132 and IDH2 R140/R172 hotspots	Caris Life Sciences (Caris Assure)	Fallback route to the IDH genotype when tissue is inadequate or exhausted.	test info · 3600 W Royal Ln, Irving, TX 75063 · 888-979-8669
Plasma ctDNA NGS panel covering IDH1 R132 and IDH2 R140/R172 hotspots	NeoGenomics Laboratories (Liquid biopsy panel)	Fallback route to the IDH genotype when tissue is inadequate or exhausted.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 239-768-0600 option 3
CD3 and CD8 tumour-infiltrating lymphocyte density plus tumour HLA class I expression by IHC, read on the dedifferentiated component	NeoGenomics Laboratories (preferred) (CD3 / CD8 immunohistochemistry)	Context for checkpoint and cell-therapy discussions; not an eligibility criterion anywhere.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 239-768-0600 option 3

CD3 and CD8 tumour-infiltrating lymphocyte density plus tumour HLA class I expression by IHC, read on the dedifferentiated component	Mayo Clinic Laboratories (T-cell subset immunohistochemistry)	Context for checkpoint and cell-therapy discussions; not an eligibility criterion anywhere.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
CD3 and CD8 tumour-infiltrating lymphocyte density plus tumour HLA class I expression by IHC, read on the dedifferentiated component	ARUP Laboratories (CD3 / CD8 immunohistochemistry)	Context for checkpoint and cell-therapy discussions; not an eligibility criterion anywhere.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 800-522-2787
Germline cancer-predisposition panel from blood or saliva, reflexed if the tumour panel returns a BRCA1, BRCA2, PALB2 or TP53 variant at an allele fraction suggesting germline origin	Myriad Genetics (preferred) (MyRisk Hereditary Cancer Test)	Family screening, and radiotherapy planning if a germline TP53 variant is confirmed.	test info · Salt Lake City, UT
Germline cancer-predisposition panel from blood or saliva, reflexed if the tumour panel returns a BRCA1, BRCA2, PALB2 or TP53 variant at an allele fraction suggesting germline origin	Ambry Genetics (CustomNext-Cancer)	Family screening, and radiotherapy planning if a germline TP53 variant is confirmed.	test info · Aliso Viejo, CA
Germline cancer-predisposition panel from blood or saliva, reflexed if the tumour panel returns a BRCA1, BRCA2, PALB2 or TP53 variant at an allele fraction suggesting germline origin	GeneDx (Hereditary cancer panel)	Family screening, and radiotherapy planning if a germline TP53 variant is confirmed.	test info · Stamford, CT
Germline cancer-predisposition panel from blood or saliva, reflexed if the tumour panel returns a BRCA1, BRCA2, PALB2 or TP53 variant at an allele fraction suggesting germline origin	Labcorp Oncology (Hereditary cancer panel)	Family screening, and radiotherapy planning if a germline TP53 variant is confirmed.	test info · 358 South Main Street, Burlington, NC 27215
Germline cancer-predisposition panel from blood or saliva, reflexed if the tumour panel returns a BRCA1, BRCA2, PALB2 or TP53 variant at an allele fraction suggesting germline origin	Mayo Clinic Laboratories (Hereditary cancer multi-gene panel)	Family screening, and radiotherapy planning if a germline TP53 variant is confirmed.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
NY-ESO-1 expression by IHC or RT-PCR, reflexed only from a documented HLA-A*02:01 result	Caris Life Sciences (preferred) (MI Cancer Seek whole transcriptome (CTAG1B expression))	NY-ESO-1-directed TCR-T trial referral (Ietetresgene autoleucel); trial-only, no approved product.	test info · 3600 W Royal Ln, Irving, TX 75063 · 888-979-8669

NY-ESO-1 expression by IHC or RT-PCR, reflexed only from a documented HLA-A*02:01 result	Tempus AI (Tempus xR RNA expression)	NY-ESO-1-directed TCR-T trial referral (letetresgene autoleucel); trial-only, no approved product.	test info · 600 West Chicago Avenue, Chicago, IL 60654
NY-ESO-1 expression by IHC or RT-PCR, reflexed only from a documented HLA-A*02:01 result	NeoGenomics Laboratories (Custom immunohistochemistry)	NY-ESO-1-directed TCR-T trial referral (letetresgene autoleucel); trial-only, no approved product.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 239-768-0600 option 3
PRAME expression by IHC or RT-PCR on the dedifferentiated component, reflexed only from a documented HLA-A*02:01 result	Caris Life Sciences (preferred) (MI Cancer Seek whole transcriptome (PRAME expression))	PRAME-directed ImmTAC or TCR-T trial referral (brenetafusp, IMA203); trial-only.	test info · 3600 W Royal Ln, Irving, TX 75063 · 888-979-8669
PRAME expression by IHC or RT-PCR on the dedifferentiated component, reflexed only from a documented HLA-A*02:01 result	Tempus AI (Tempus xR RNA expression)	PRAME-directed ImmTAC or TCR-T trial referral (brenetafusp, IMA203); trial-only.	test info · 600 West Chicago Avenue, Chicago, IL 60654
PRAME expression by IHC or RT-PCR on the dedifferentiated component, reflexed only from a documented HLA-A*02:01 result	NeoGenomics Laboratories (PRAME immunohistochemistry)	PRAME-directed ImmTAC or TCR-T trial referral (brenetafusp, IMA203); trial-only.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 239-768-0600 option 3

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements
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IDH1 codon 132 and IDH2 codons 140/172 mutation testing by DNA sequencing (comprehensive NGS panel or targeted hotspot assay), not R132H	Sequencing decides whether an IDH-directed drug or trial is available at all, and it is the highest-yield untested item in the case: IDH1/IDH2 mutations were present in 71 percent of dedifferentiated chondrosarcomas in a paired series against 36 percent of conventional tumours (pmid:36926116). Order sequencing rather than R132H immunohistochemistry; the R132H antibody was raised for glioma, and roughly 40 percent of mutant cartilage tumours carry R132C, with R132G, R132L and R132S making up much of the rest (pmid:21598255), so a stain would read negative in the commonest scenario here. Skipping the test leaves the one established targeted handle in chondrosarcoma unassessable rather than excluded.	Foundation Medicine (FoundationOne CDx) · test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639	Archival FFPE block or 10-15 unstained slides cut from the dedifferentiated component
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Expert bone-sarcoma pathology review of the diagnostic material, reporting the WHO diagnosis and the percentage of the specimen occupied by the dedifferentiated (non-cartilaginous) component	The histology is on file as user-stated, and it is the fact that makes cytotoxic chemotherapy worth giving: dedifferentiated and mesenchymal are the only two potentially chemosensitive chondrosarcoma subtypes, with first-line ORR around 20 percent and median PFS of 4-5 months in advanced disease, while conventional, clear-cell and periosteal disease is chemoresistant (pmid:41895355). Reference-centre review also fixes the diagnosis that every trial protocol screens on (pmid:34500044), and the size of the dedifferentiated fraction is what the chemotherapy expectation rests on. Without it, an anthracycline regimen may be given for a tumour that will not respond, or withheld from one that would.	Mayo Clinic Laboratories (Consultative bone and soft tissue pathology) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710	Diagnostic H&E slides plus the paraffin block; no new biopsy
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Comprehensive tumour DNA plus RNA next-generation sequencing panel covering IDH1/IDH2, TP53, CDKN2A/B, TERT promoter, NTRK1-3, RET and BRAF, reporting TMB in mutations per megabase and an MSI call	One order answers the IDH question and every tumour-agnostic one at the same time, which is what a patient with no molecular testing needs. Dedifferentiated chondrosarcoma is enriched for TP53, TERT promoter and CDKN2A/B alterations relative to conventional disease (pmid:36926116), while TMB of 10 or more per megabase and MSI-high are rare in sarcoma at 3.9 percent and 0.7 percent respectively (pmid:35705558, pmid:42531431) but each opens approved checkpoint therapy. Specify a validated comprehensive panel with RNA fusion coverage; a small hotspot panel prints a TMB figure it is not sized to call, and a DNA-only panel can miss a fusion whose breakpoint sits in a large intron.	Foundation Medicine (FoundationOne CDx) · test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639	One FFPE block or 10-20 unstained slides containing the dedifferentiated component
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Pathology block review for tumour cellularity and representation, with macrodissection instructions and a written tissue-allocation plan across the panel, the set and any reflex stains	Everything else on this list runs off one block, and dedifferentiated chondrosarcoma is biphasic by definition, so a core that sampled only the low-grade cartilage returns a falsely reassuring molecular and PD-L1 result; PD-L1 positivity in this subtype was confined to the dedifferentiated component (pmid:27312065), and the two components differ in mutational and copy-number profile even though they share early events (pmid:36926116). Ask the pathologist to confirm the block holds enough high-grade non-cartilaginous tumour, to macrodissect it, and to cut once for the panel and once for the stains rather than piecemeal. If the block falls short, schedule a repeat image-guided core of the acetabular mass now instead of waiting for a stabilisation procedure.	Treating centre pathology service (the review happens where the diagnostic block lives; no send-out)	Existing diagnostic block; a repeat image-guided core only if tumour content is insufficient
Documented ECOG performance status scored in clinic, recorded alongside weight-bearing status, ambulatory aids and analgesic requirement	The ECOG of 2 in the profile was assumed at intake from the pathologic fracture and has never been measured, yet it carries more weight than any other number in the case: most protocols cap eligibility at 0-1 or 0-2, and full-dose anthracycline plus ifosfamide is a fitness decision. A score driven by a mechanically unstable hip is not the same as one driven by disease burden, and stabilisation or adequate analgesia may move it. Until someone scores it, every eligibility statement made downstream rests on a guess.	Treating oncology clinic (scored at the next visit; no send-out)	None; clinic assessment

Baseline transthoracic echocardiogram with LVEF (or MUGA), plus a 12-lead ECG with QTc	Anthracycline-based chemotherapy is the expected first-line treatment in this histology (pmid:41895355), which turns a baseline LVEF from a formality into a prerequisite, and the phase 3 ivosidenib trial excludes an LVEF below 40 percent measured within 28 days of randomisation (nct:NCT06127407). Ivosidenib prolongs the QT interval, so the same visit should capture an ECG and make later IDH-directed screening cheaper. None of this waits on the molecular results.	Treating centre cardiology and echocardiography service (no send-out)	None; echocardiogram and ECG
Orthopaedic oncology assessment of the periacetabular fracture with thin-slice bone-algorithm pelvic CT and MRI of the hemipelvis, addressing stabilisation or resection options and their timing against systemic therapy	The hardening question at this site is structural rather than molecular: whether the joint needs fixation or resection, and whether that happens before, alongside or after chemotherapy. A primary pelvic sarcoma with a pathologic fracture is a different surgical problem from an acetabular metastasis, since fracture haematoma contaminates compartments and periacetabular resection and reconstruction is a specialist decision (pmid:41289179, pmid:34500044). The sequencing carries a concrete cost: the phase 3 ivosidenib protocol excludes major surgery within 4 weeks of randomisation (nct:NCT06127407), and comparable washouts are standard, so an unplanned operation can close trial doors for a month.	Orthopaedic oncology service at a high-volume sarcoma centre, with the treating radiology department	None; imaging plus clinical review

HLA class I genotyping at 4-digit (allele-level) resolution by sequence-based typing	A blood draw that spends no tumour tissue and decides whether the three cancer-testis antigen rows below are worth staining at all. Every TCR-T and ImmTAC programme in play names a restricting allele, most often HLA-A02:01, and afamitresgene autoleucel was licensed on exactly that pairing in synovial sarcoma (pmid:38554725). Order it now: a non-A02 type closes the whole class in one step and preserves the block, while typing later adds several weeks to any cell-therapy referral.	Mayo Clinic Laboratories (HLA class I high-resolution typing) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710	5-10 mL whole blood in EDTA, or a buccal swab; no tumour tissue
Four-antibody mismatch-repair IHC (MLH1, PMS2, MSH2, MSH6), with MSI by PCR or NGS if any stain is equivocal	Days rather than weeks, and an orthogonal read on the MSI call the sequencing panel will produce. MSI-high appeared in 0.7 percent of sarcomas in the largest clinically sequenced cohort (pmid:42531431), so a proficient result is the expected one, but a loss of staining would put approved tumour-agnostic checkpoint blockade in front of a patient with no other approved targeted option. Loss of a mismatch-repair protein also raises a Lynch syndrome question for her and her relatives, and reflexes to MLH1 promoter methylation testing.	NeoGenomics Laboratories (MMR IHC panel) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 239-768-0600 option 3	4 unstained slides from the block cut for the immunohistochemistry set

Baseline contrast-enhanced CT of chest, abdomen and pelvis with thin-section lung reconstructions, read for RECIST 1.1 measurability and for whether the nodule distribution leaves any role for local therapy	Two questions get settled in one acquisition: whether any nodule reaches 10 mm in longest diameter and is therefore measurable under RECIST 1.1 (pmid:19097774), which trial protocols including the phase 3 ivosidenib study require, and whether bilateral dispersed disease genuinely rules out metastasectomy or SBRT. Small scattered nodules are also the yardstick against which chemotherapy response will be judged, so the baseline has to exist before treatment starts rather than after the first cycle. Biopsy of a pulmonary nodule is worth doing only if the plan turns on tissue confirmation of metastasis or on the metastatic genotype.	Treating centre radiology service (no send-out)	None; imaging
Baseline organ-function panel: CBC with differential, comprehensive metabolic panel, measured or calculated creatinine clearance, and urinalysis for proteinuria and tubular function before ifosfamide or cisplatin	Renal and hepatic panels are both recorded as not obtained, and the regimens actually in play for dedifferentiated disease (doxorubicin plus ifosfamide, doxorubicin plus cisplatin, MAP-type schedules) each carry organ-specific dose limits, with ifosfamide tubular toxicity and cisplatin nephrotoxicity the usual constraints (pmid:41895355). Trial screening will ask for the same values within a defined window, so drawing them at the same visit as the echocardiogram avoids a repeat trip.	Treating centre or any accredited hospital laboratory (no send-out)	Routine blood draw and a urine sample

Ancillary subtyping to separate dedifferentiated chondrosarcoma from chondroblastic osteosarcoma, mesenchymal chondrosarcoma and grade 3 conventional chondrosarcoma: morphology review plus RNA fusion testing for HEY1-NCOA2 and the IDH1/IDH2 genotype	The differential changes the chemotherapy backbone: chondroblastic osteosarcoma points toward a MAP-type regimen, mesenchymal chondrosarcoma is HEY1-NCOA2 driven and treated on Ewing-like lines, and dedifferentiated disease is usually given doxorubicin with ifosfamide or cisplatin (pmid:41895355). Genotype helps arbitrate, since IDH1/IDH2 mutations were confined to central and periosteal cartilaginous tumours and absent from 222 osteosarcomas in the founding series (pmid:21598255). Both readouts come off the same panel already being ordered, so the marginal cost is a line on the requisition.	Caris Life Sciences (MI Cancer Seek whole transcriptome) · test info · 3600 W Royal Ln, Irving, TX 75063 · 888-979-8669	Same FFPE block; no additional tissue when the fusion panel is ordered with the DNA panel
IDH1/IDH2 sequencing performed on macrodissected dedifferentiated tumour, with the sampled lesion documented on the report; consider a metastatic-site sample if an IDH-directed therapy is being pursued	Mutant IDH1 is recurrently lost during the evolution of metastatic central chondrosarcoma, which the authors flag as a direct problem for treating metastatic disease with IDH inhibitors (pmid:41299743). A mutation called on the acetabular primary therefore does not guarantee that the pulmonary metastases still carry it, and the two components of a dedifferentiated tumour need to be handled separately even where they share early events (pmid:36926116). Ask for macrodissection of the high-grade component and for the sampled site to appear on the report, so the result can be read for what it actually covers.	Foundation Medicine (FoundationOne CDx) · test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639	Same archival block, macrodissected; a lung nodule biopsy only if an IDH-directed treatment decision hinges on the metastatic genotype

PD-L1 IHC with the clone and scoring system recorded explicitly (22C3 pharmDx with CPS, or SP263 per the protocol in play), read on a section containing the dedifferentiated component	This is a subtype-specific finding rather than a generic sarcoma prior: PD-L1 was positive in 9 of 22 dedifferentiated chondrosarcomas and in none of 119 conventional, 19 mesenchymal or 20 clear-cell tumours, with staining confined to the dedifferentiated component and tracking with T-cell infiltrate (pmid:27312065); a second cohort found comparable positivity in dedifferentiated disease (pmid:33912442). No sarcoma indication is keyed to PD-L1, and SARC028 showed limited activity in bone sarcoma cohorts (pmid:28988646), so treat the result as trial-enrichment information rather than a prescription. Stain a section that actually contains dedifferentiated tissue, or the result will be falsely negative for the wrong reason.	NeoGenomics Laboratories (PD-L1 IHC (22C3, SP263, 28-8)) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 239-768-0600 option 3	1-2 unstained slides from the immunohistochemistry-set block
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MAGE-A4 expression by IHC or RT-PCR on the dedifferentiated component, ordered only after HLA-A*02:01 is documented; RNA expression from a panel is an acceptable first pass	The sequence is what makes this tractable: type first, stain only if HLA-A*02:01 is present, because an expression result cannot be interpreted for eligibility without the restricting allele. Expectations should be modest; in the largest cancer-testis antigen survey of bone and soft tissue sarcomas MAGE-A4 was found in about 40 percent of osteosarcomas and a third of undifferentiated pleomorphic sarcomas, while chondrosarcomas showed little expression overall (pmid:40152485), and the approved product is licensed for synovial sarcoma alone (pmid:38554725). Sponsors re-test centrally with their own antibody and cutoff, so a local result triages the referral rather than qualifying her for it.	Caris Life Sciences (MI Cancer Seek whole transcriptome (MAGEA4 expression)) · test info · 3600 W Royal Ln, Irving, TX 75063 · 888-979-8669	1-2 unstained slides from the immunohistochemistry-set block; reflex only
Plasma ctDNA NGS panel covering IDH1 R132 and IDH2 R140/R172 hotspots	A backstop for the IDH call when the block is exhausted or holds too little dedifferentiated tumour, and the shedding data are encouraging in exactly this population: circulating tumour DNA was detected by digital PCR in all six patients with grade 3 or dedifferentiated chondrosarcoma, and never in grade 1 disease (pmid:28834325). A negative plasma result does not exclude the mutation, so tissue stays the reference standard and a blood-only negative should not be used to close the IDH question.	Guardant Health (Guardant360 CDx) · test info · 3100 Hanover Street, Palo Alto, CA 94304	2 Streck tubes of whole blood (roughly 20 mL); no tissue

CD3 and CD8 tumour-infiltrating lymphocyte density plus tumour HLA class I expression by IHC, read on the dedifferentiated component	Dedifferentiated chondrosarcoma carries T-cell infiltrate and retained HLA class I far more often than the other subtypes, and PD-L1 positivity travels with both (pmid:27312065, pmid:33912442). Loss of tumour HLA class I would undercut any T-cell-redirecting approach even with the right host genotype and a positive antigen stain, which is the one thing the HLA blood draw cannot tell you. No protocol enrolls on these numbers, so they belong on the same block cut as the other stains and not on a separate requisition.	NeoGenomics Laboratories (CD3 / CD8 immunohistochemistry) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 239-768-0600 option 3	2-3 unstained slides from the immunohistochemistry-set block
Germline panel from blood or saliva, reflexed if the tumour panel returns a BRCA1, BRCA2, PALB2 or TP53 variant at an allele fraction suggesting germline origin	The cartilage phenotype itself does not point to a germline syndrome: no germline IDH1 or IDH2 mutations were found across roughly 1200 mesenchymal tumours in the founding series, and the multiple-enchondroma syndromes are mosaic rather than inherited (pmid:21598255). The germline question here therefore rides on what else the tumour panel turns up, and deficiency appeared in 2.5 percent of sarcomas in a large profiling cohort (pmid:35705558). A confirmed germline BRCA1/2 or TP53 finding changes screening for her relatives and, in the TP53 case, how radiotherapy is considered.	Myriad Genetics (MyRisk Hereditary Cancer Test) · test info · Salt Lake City, UT	5-10 mL whole blood in EDTA, or a saliva collection kit

NY-ESO-1 expression by IHC or RT-PCR, reflexed only from a documented HLA-A*02:01 result	Unassessable until the HLA type exists, and low yield if the type comes back A02:01: NY-ESO-1 expression was low across every sarcoma subtype in the largest cancer-testis antigen survey of bone and soft tissue tumours (pmid:40152485), and the clinical programmes remain confined to synovial sarcoma and myxoid liposarcoma. Include it as a reflex on the same block cut as the other stains, and drop it outright if the type is not HLA-A02:01.	Caris Life Sciences (MI Cancer Seek whole transcriptome (CTAG1B expression)) · test info · 3600 W Royal Ln, Irving, TX 75063 · 888-979-8669	1 unstained slide from the immunohistochemistry-set block; reflex only
PRAME expression by IHC or RT-PCR on the dedifferentiated component, reflexed only from a documented HLA-A*02:01 result	Cannot be called without the HLA type, and the prior is unfavourable: of 13 chondrosarcomas stained in a multi-subtype sarcoma series, 1 was PRAME-positive, against 48 percent of undifferentiated pleomorphic sarcomas (pmid:40152485). Everything behind the target is trial-only, which puts this at the bottom of the list even though the reflex stain itself is cheap. Read it on the dedifferentiated component if it is ordered at all.	Caris Life Sciences (MI Cancer Seek whole transcriptome (PRAME expression)) · test info · 3600 W Royal Ln, Irving, TX 75063 · 888-979-8669	1 unstained slide from the immunohistochemistry-set block; reflex only

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.